Implementation Document

Topic 14: Switch Node & Workflow Routing

1. Workflow Overview

The workflow uses n8n built-in nodes to route user profiles according to their roles.

Workflow Nodes

1. Manual Trigger
2. Mock Users
3. Split Users
4. Route by Role
5. Set Admin
6. Set Editor
7. Set Viewer
8. Set Guest

2. Node Configuration

2.1 Manual Trigger

Node: When clicking "Test workflow"

Purpose:

- Starts the workflow manually.
- Allows the trainee to test the complete workflow.

2.2 Mock Users

Node: Edit Fields (Set)

Configuration:

- Node Name: Mock Users
- Mode: JSON

Input:

```
None
{
  "profiles": [
    {"name": "Alex", "role": "admin"},
    {"name": "Sarah", "role": "editor"},
    {"name": "John", "role": "viewer"},
    {"name": "David", "role": "guest"}
  ]
}
```

Purpose:

- Creates four mock user profiles.
- Supplies sample data for the routing process.

2.3 Split Users

Node: Split Out

Configuration:

- Node Name: Split Users
- Fields To Split Out: `profiles`
- Include: No Other Fields

Purpose:

- Splits the profiles array into individual workflow items.
- Allows each user to be evaluated separately.

Expected items:

- Alex — admin
- Sarah — editor
- John — viewer
- David — guest

2.4 Route by Role

Node: Switch

Configuration:

- Node Name: Route by Role
- Mode: Rules
- Value 1: `{{ $json.role }}`
- Data Type: String, where available
- Operation: is equal to
- Fallback Output: Extra Output

Routing Rules

Rule 1

- Value 2: `admin`
- Output: 0

Rule 2

- Value 2: `editor`
- Output: 1

Rule 3

- Value 2: `viewer`
- Output: 2

Fallback

- Output: 3

Purpose:

- Evaluates the role field.
- Routes each item to its matching output.
- Sends unmatched roles to the fallback output.

2.5 Set Admin

Node: Edit Fields (Set)

Configuration:

Field	Type	Value
status	String	Admin Access

Connection:

- Route by Role output 0 → Set Admin

2.6 Set Editor

Node: Edit Fields (Set)

Configuration:

Field	Type	Value
status	String	Editor Access

Connection:

- Route by Role output 1 → Set Editor

2.7 Set Viewer

Node: Edit Fields (Set)

Configuration:

Field	Type	Value
status	String	Viewer Access

Connection:

- Route by Role output 2 → Set Viewer

2.8 Set Guest

Node: Edit Fields (Set)

Configuration:

Field	Type	Value
status	String	Guest Access

Connection:

- Route by Role output 3 → Set Guest

Important Configuration

Enable **Include Other Input Fields** in each downstream Edit Fields node to preserve the original user information.

3. Execution Procedure

1. Open the Topic 14 workflow.
2. Verify the Mock Users input.
3. Execute the workflow.
4. Confirm that Split Users generates four items.
5. Verify the Switch node routes the items.
6. Inspect each downstream Edit Fields node.
7. Confirm that the correct status field is added.
8. Save the workflow.

4. Expected Output

Example for the Admin branch:

```
None
[
  {
    "name": "Alex",
    "role": "admin",
    "status": "Admin Access"
  }
]
```

Example for the Editor branch:

```
None
[
  {
    "name": "Sarah",
    "role": "editor",
    "status": "Editor Access"
  }
]
```

Example for the Viewer branch:

```
None
[
  {
    "name": "John",
    "role": "viewer",
    "status": "Viewer Access"
  }
]
```

Example for the Guest branch:

```
None
[
  {
    "name": "David",
    "role": "guest",
    "status": "Guest Access"
  }
]
```

5. Conclusion

The implementation demonstrates how the Switch node can route multiple JSON items using role-based conditions and a fallback output. The four downstream nodes add the appropriate status while preserving the original user fields.